

Ranking & Ordering

COMPLETE LEARNING GUIDE

What You'll Learn

Ranking and ordering questions establish a linear order among people or items (by rank in class, height, age, marks, or arrival order) using a mix of absolute positions, relative comparisons, and 'from-both-ends' clues, then ask for a specific rank, a total count, or a comparison between two people. Unlike full seating puzzles, these are usually solvable with a couple of core formulas plus careful bookkeeping of who is compared to whom, rather than a full grid. The two recurring skills are converting between 'rank from top' and 'rank from bottom' using the total-count formula, and correctly chaining comparative statements ('taller than', 'scored more than') into a single ordered list. This sub-module covers the from-both-ends formula, total-count-from-two-ranks problems, chaining comparative clues, handling ties and 'just above/below' language, combining ranking with numeric gaps, and common group/sub-group ranking scenarios.

1. The From-Both-Ends Formula

For a person ranked at position R from the top in a group of total size N , their rank from the bottom is $(N - R + 1)$. This single formula underlies almost every 'rank from top vs rank from bottom' question, and the safest approach is to always convert every given rank to 'rank from top' before comparing people, so you're never mixing two different reference directions.

- In a class of 40 students, Meena is ranked 12th from the top. Her rank from the bottom = $40 - 12 + 1 = 29$ th. Conversely, if a rank from the bottom is given as 29th, rank from top = $40 - 29 + 1 = 12$ th — the formula is symmetric.

2. Finding Total Count From Two Given Ranks

When a single person's rank from the top and rank from the bottom are BOTH given, the total group size is found with $N = (\text{rank from top}) + (\text{rank from bottom}) - 1$, since that person is counted once but appears in both counts.

- Arjun is 15th from the top and 22nd from the bottom in his class. Total students = $15 + 22 - 1 = 36$. (The -1 avoids double-counting Arjun himself, who is included in both the 15 counted from the top and the 22 counted from the bottom.)

3. Counting Between Two Ranked People

To find how many people lie strictly between two people whose ranks (measured the same direction) are known, subtract the ranks and subtract 1 more to exclude both endpoints.

- P is 8th from the top and Q is 19th from the top (same class, same direction). Number of people strictly between them = $19 - 8 - 1 = 10$.

4. Chaining Comparative Clues Into a Single Order

When clues compare pairs ('A scored more than B', 'C is taller than A but shorter than D'), build a single ordered list incrementally, inserting each new person relative to the ones already placed rather than tracking pairwise comparisons separately — a person mentioned in multiple clues acts as the anchor connecting different parts of the chain.

- 'B scored more than A. C scored more than B but less than D. E scored less than A.' Chain from highest to lowest: $D > C > B > A > E$. (D is highest since $C < D$ and $C > B > A > E$ places everyone below C except D.)

5. Handling Ties and 'Just Above/Below' Language

'Just above' or 'immediately above' in rank means adjacent rank (no gap), exactly like 'immediately left of' in seating puzzles — a plain 'above' or 'better than' allows any gap. When a question explicitly allows ties (two people sharing a rank), treat rank as a set-based count rather than assuming every rank number is used exactly once.

- 'Rita is ranked immediately above Sam. Sam is 14th.' Since 'immediately above' means adjacent with no gap, Rita is ranked 13th (one position better, i.e., numerically lower, than 14th).

6. Ranking Combined With Numeric Gaps

Some questions give a numeric gap between ranks rather than 'immediately above' — e.g., 'there are 5 people between X and Y' — which must be converted to a rank difference of 6 (5 people between plus 1 for X or Y's own position) before use in the from-both-ends or total-count formulas.

- 'There are 5 people between X (ranked 10th from the top) and Y.' If Y is below X, Y's rank = $10 + 5 + 1 = 16$ th from the top (5 people in between, plus Y's own next position).

7. Sub-Group Ranking Scenarios

When a rank is given within a sub-group (e.g., 'rank among boys') as well as an overall rank, keep the two rank systems clearly separate on paper — do not mix a sub-group rank into an overall-group formula, since they are counted against different total sizes.

Common Pitfalls

- Mixing 'rank from top' and 'rank from bottom' values in the same calculation without converting them to one common direction first.
- Forgetting the $+ 1$ correction when computing total group size from a person's two given ranks (from top and from bottom).
- Treating 'above' or 'better than' as adjacency when only 'immediately above/below' guarantees no gap between ranks.
- Miscounting 'people between two ranks' by forgetting to subtract 1 to exclude the two endpoint ranks themselves.
- Converting a 'number of people between X and Y' clue into a rank difference without adding 1 for the endpoint's own position.
- Confusing an overall-group rank with a sub-group rank (e.g., rank among boys vs rank in the whole class) and applying one formula to both.
- Building a comparative order clue by clue without anchoring around a person common to multiple clues, causing the chain to fragment.

Key Takeaways

- Rank from bottom = $N - (\text{rank from top}) + 1$, where N is the total group size — the single formula behind most ranking questions.
- When both ranks (top and bottom) for one person are given, total count $N = \text{rank-from-top} + \text{rank-from-bottom} - 1$.
- People strictly between two same-direction ranks = $(\text{higher rank number}) - (\text{lower rank number}) - 1$.
- 'Immediately above/below' means adjacent rank (no gap); plain 'above/below' allows any gap — never conflate the two.
- Convert 'X people between A and B' to a rank difference of $X + 1$ before plugging into any formula.
- Keep sub-group ranks (e.g., among boys) and overall-group ranks in clearly separate systems — never combine them directly.

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IndiaBix — Ranking Test

<https://www.indiabix.com/logical-reasoning/ranking-test/>

GeeksforGeeks — Logical Reasoning Questions and Answers

<https://www.geeksforgeeks.org/logical-reasoning-questions-and-answers/>